Section A-

Q1) Explain the operational differences between static and dynamic routing, and justify why configuring static routes would be the most efficient choice in this scenario.

Static Routing vs Dynamic Routing

1. Operational Differences

Static Routing	**Dynamic Routing**
Routes are manually configured by the network administrator.	Routes are learned automatically using routing protocols.
Does not automatically adjust routes when the network changes.	Automatically updates routes when network changes occur.
Uses fewer CPU and memory resources.	Uses more CPU and memory resources.
Simple to configure in small networks.	Requires more configuration and protocol management.
Examples: manually configured routes using ip route.	Examples: RIP, OSPF, and EIGRP.

Why Static Routing Is the Best Choice

In this startup scenario, there are only two routers connecting two offices, and the network topology rarely changes. Therefore, static routing is the most efficient choice for the following reasons:

1. Simple configuration: The administrator can manually configure routes between the two offices.
2. Low resource usage: Static routing consumes less CPU, memory, and network bandwidth because it does not exchange routing updates.
3. Easy maintenance: With only two routers and a stable topology, maintaining static routes is straightforward.
4. Better control: The administrator has direct control over the paths used for network traffic.
5. Cost-effective: No complex routing protocol configuration or monitoring is required.

Q2) Compare RIP and OSPF as dynamic routing protocol options for this network, and recommend which one is better suited as the network grows, justifying your choice based on convergence time and scalability.

Comparison of RIP and OSPF

As the startup expands to five branch routers, dynamic routing becomes more suitable because it automatically learns routes and reduces manual configuration errors.

1. RIP vs OSPF

Feature	**RIP**	**OSPF**
Full form	Routing Information Protocol	Open Shortest Path First
Routing method	Distance-vector	Link-state

Metric	Hop count	Cost based on link bandwidth
Maximum hops	15 hops; 16 means unreachable	No fixed 15-hop limit
Convergence time	Generally slower	Generally faster
Scalability	Best for small networks	Suitable for medium and large networks
Updates	Periodic routing updates	Link-state updates when topology changes
Configuration	Simple	More complex

2. Recommended Protocol: OSPF

OSPF is the better choice for the growing startup network for these reasons:

1. Faster convergence: OSPF generally detects network changes and recalculates routes faster than RIP, helping restore connectivity quickly after a link failure.
2. Better scalability: OSPF can support network growth beyond five routers without RIP's 15-hop limitation.
3. Efficient routing: OSPF selects routes using link cost, typically based on bandwidth, rather than simply counting hops.
4. Reduced manual work: Routers exchange routing information automatically, reducing the need to configure every remote route manually.

Q-3) Explain how the switch populates its MAC address table and forwards frames between the two devices, including what happens before the table is populated.

1. How the Switch Works

1. Receiving a frame: When Device A sends a frame, the switch receives it on a port.
2. Learning the MAC address: The switch reads the source MAC address of the frame and records it in its MAC address table along with the incoming port.
3. Checking the destination: The switch checks its MAC address table to find the destination MAC address of Device B.
4. Unknown destination: If Device B's MAC address is not in the table, the switch floods the frame out of all other ports in the same VLAN, except the incoming port.
5. Learning Device B: When Device B replies, the switch learns Device B's source MAC address and records its port.
6. Forwarding frames: Once both MAC addresses are learned, the switch forwards subsequent frames only through the appropriate destination port.

2. Example MAC Address Table

MAC Address	Switch Port	Device
AA:AA:AA:AA:AA:AA	Fa0/1	Device A
BB:BB:BB:BB:BB:BB	Fa0/2	Device B

3. What Happens Before the Table Is Populated?

Initially, the switch's MAC address table may be empty. When Device A sends a frame to Device B, the switch learns Device A's source MAC address but does not yet know Device B's port.

Therefore, it floods the frame to other ports in the same VLAN. After Device B replies, the switch learns its MAC address and can forward future frames directly.

Q-4) Explain the purpose of VLAN segmentation, and describe how an 802.1Q trunk link allows traffic for multiple VLANs to travel between the switch and the router over a single physical connection

VLAN Segmentation and 802.1Q Trunking

1. Purpose of VLAN Segmentation

A Virtual Local Area Network (VLAN) logically divides a physical switch into separate broadcast domains. It allows the Marketing and Engineering teams to use the same physical switch while keeping their network traffic logically separated.

Benefits of VLAN segmentation:

1. Security: Separates Marketing and Engineering traffic.
2. Reduced broadcast traffic: Broadcasts in one VLAN do not normally reach devices in another VLAN.
3. Better management: Each department can have its own network and IP subnet.
4. Improved performance: Reduces unnecessary broadcast traffic within each VLAN.

Example:

Department	VLAN ID	Network
Marketing	10	192.168.30.0/24
Engineering	20	192.168.40.0/24

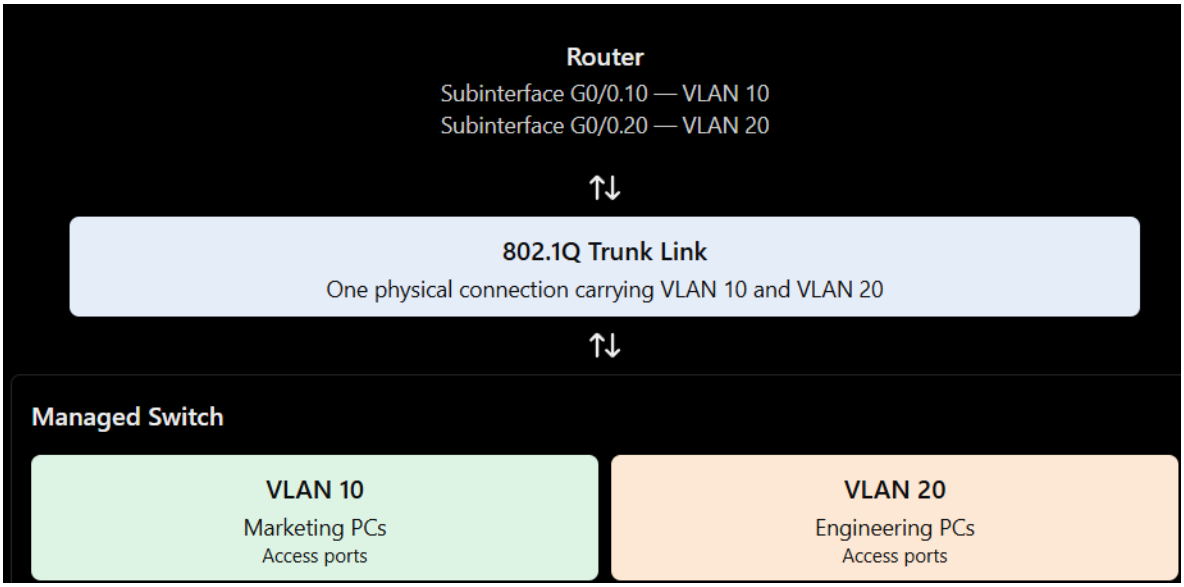
2. How an 802.1Q Trunk Link Works

An 802.1Q trunk link carries traffic from multiple VLANs over a single physical connection between a switch and a router.

- The switch assigns connected devices to VLANs using access ports.
- When traffic travels over the trunk, the switch adds an 802.1Q VLAN tag to identify the VLAN associated with each frame.
- The router receives the tagged frames through one physical interface configured with separate logical subinterfaces.
- Each subinterface handles traffic for its assigned VLAN.
- The router can route traffic between VLANs when appropriate routing and security policies permit it.

This setup is commonly called Router-on-a-Stick.

3. Simple Network Diagram



Q-5)

Question: Explain why inter-VLAN routing is required here, and justify the specific advantages of a Router-on-a-Stick configuration over adding a second physical router interface.

Inter-VLAN Routing

In this scenario, the startup has two VLANs:

- **Marketing VLAN** → e.g. VLAN 10
- **Engineering VLAN** → e.g. VLAN 20
- Both VLANs are connected to the **same switch**
- There is only **one router**

Why is Inter-VLAN Routing required?

By default, **different VLANs are separate Layer-2 broadcast domains**.

For example:

Marketing VLAN 10 Engineering VLAN 20

PC1 ——————┐ ┌—————— PC3

| |

[Switch]

|

VLAN 10 ≠ VLAN 20

A normal Layer-2 switch can forward frames **within the same VLAN**, but it cannot route traffic from VLAN 10 to VLAN 20.

So if:

PC1 (VLAN 10)

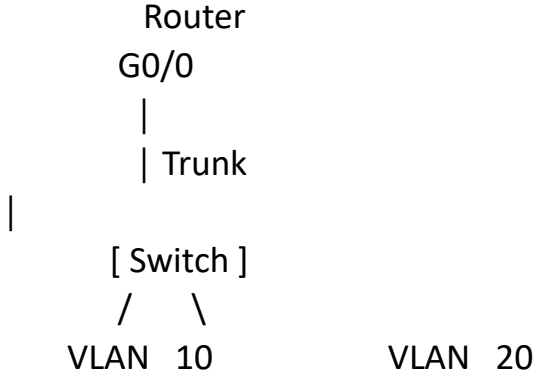
IP: 192.168.10.10

↓

PC3 (VLAN 20)
IP: 192.168.20.10
PC1 needs a **Layer-3 device** to move the packet between the two different networks.
That is the job of **inter-VLAN routing**.
Why Router-on-a-Stick?

Instead of adding another physical router interface, we can use **one physical router interface divided into multiple logical subinterfaces**.

Example:



On the router:
G0/0.10 → VLAN 10 → 192.168.10.1
G0/0.20 → VLAN 20 → 192.168.20.1

The switch sends VLAN-tagged traffic through the single trunk link, and the router's subinterfaces route traffic between the VLANs.
Main advantages

1. **Saves hardware**

You don't need a separate physical router interface for every VLAN.
Traditional:
VLAN 10 → Physical Interface
VLAN 20 → Physical Interface
VLAN 30 → Physical Interface

With Router-on-a-Stick:
VLAN 10 ─┐
VLAN 20 ─┼──→ ONE physical router interface
VLAN 30 ─┘

2. **Lower cost**

For a small startup, using one existing router avoids buying additional router hardware or interface modules.

3. **Easy to expand**

If the company adds another VLAN, you can create another subinterface rather than installing another physical interface.
Example:
G0/0.10 → Marketing
G0/0.20 → Engineering
G0/0.30 → HR
G0/0.40 → Finance

4. **Centralized configuration**

All inter-VLAN routing can be configured on the same router.
Router-on-a-Stick vs Multiple Physical Interfaces

Router-on-a-Stick	**Multiple physical interfaces**
One physical interface	One physical interface per VLAN
Uses subinterfaces	Uses physical interfaces
Lower hardware cost	More hardware required
Easy to add VLANs	Limited by available interfaces
Requires trunk link	Separate links/interfaces
Suitable for small networks	Can provide better performance at higher traffic

One important disadvantage

Router-on-a-Stick has **one physical link carrying traffic for multiple VLANs**. Therefore, that link can become a bottleneck if the network has heavy inter-VLAN traffic.
For this **small startup with a single router**, however, Router-on-a-Stick provides a practical way to enable Marketing ↔ Engineering communication without adding another physical router interface.
"Inter-VLAN routing is required because VLANs are separate Layer-2 broadcast domains and a Layer-2 switch cannot route traffic between them. Router-on-a-Stick allows one physical router interface to provide routing for multiple VLANs using logical subinterfaces over a trunk link, reducing hardware cost and making the network easier to scale."

Question-6 : Explain the wireless security and MAC filtering options available, and justify which configuration approach best prevents unauthorized access while still allowing convenient onboarding of new employee devices.

Wireless Security & MAC Filtering — Easy Explanation

In this scenario, the startup has an **open-plan office** where unauthorized devices have connected before

Only employees should be able to access the Wi-Fi, while adding a new employee device should remain easy.

1. Wireless Security Options

Security	How it works	Security level
Open	No password	❌ Very poor
WEP	Old encryption method	❌ Insecure/obsolete
WPA/WPA2-Personal	Shared Wi-Fi password	☑ Good
WPA3-Personal	Modern password-based security	☑☑ Strong
WPA2/WPA3-Enterprise	Individual employee authentication through authentication server	☑☑☑ Best for organizations

For a startup, **WPA3-Personal** is a strong choice if all employee devices support it. If compatibility is an issue, **WPA2/WPA3 mixed mode** can be considered.

Every network device has a MAC address.

Example:

Employee Laptop
MAC: AA:BB:CC:11:22:33

The access point can maintain a list of allowed MAC addresses. **Allow-list**

Allowed:
AA:BB:CC:11:22:33 → Laptop 1
AA:BB:CC:44:55:66 → Laptop 2

AA:BB:CC:77:88:99 → Phone

Only devices on the list are allowed to connect.
Deny-list

You can also specifically block known devices.

Blocked:
XX:XX:XX:XX:XX:XX

Best Configuration for This Scenario

For a startup that wants **security + convenient employee onboarding**, I would use: **WPA2/WPA3-Enterprise + individual employee authentication**

Ideally:

Employee Device
↓
Wi-Fi AP
↓
WPA2/WPA3-Enterprise
↓
Authentication Server
↓
Employee credentials
↓
Access

Each employee can have their **own credentials**, rather than everyone sharing one Wi-Fi password.

Why this is convenient

When a new employee joins:

New Employee
↓
Create employee account
↓
Configure Wi-Fi on laptop/phone
↓
Authenticate
↓
Access granted

When an employee leaves:

Employee leaves
↓
Disable their account
↓
Their Wi-Fi access stops

You don't need to change the Wi-Fi password for the entire company.

If the startup specifically wants MAC filtering, use it as an **additional layer**, preferably with an **allowlist**, rather than relying on it as the main security mechanism.

However, maintaining a MAC allow-list for every employee device can make onboarding and device replacement more inconvenient.

So the practical priority is:

Strong authentication first → MAC filtering as optional additional control.

"Open Wi-Fi and WEP should be avoided because they provide inadequate security. WPA2/WPA3 provides encrypted access, while WPA2/WPA3-Enterprise allows each employee to authenticate individually. MAC filtering can allow or block specific devices, but it should not be relied upon as the primary security mechanism because MAC addresses can be spoofed. For a startup, WPA2/WPA3-Enterprise with individual employee authentication provides strong security and convenient onboarding because new employees can be added without changing a shared Wi-Fi password."